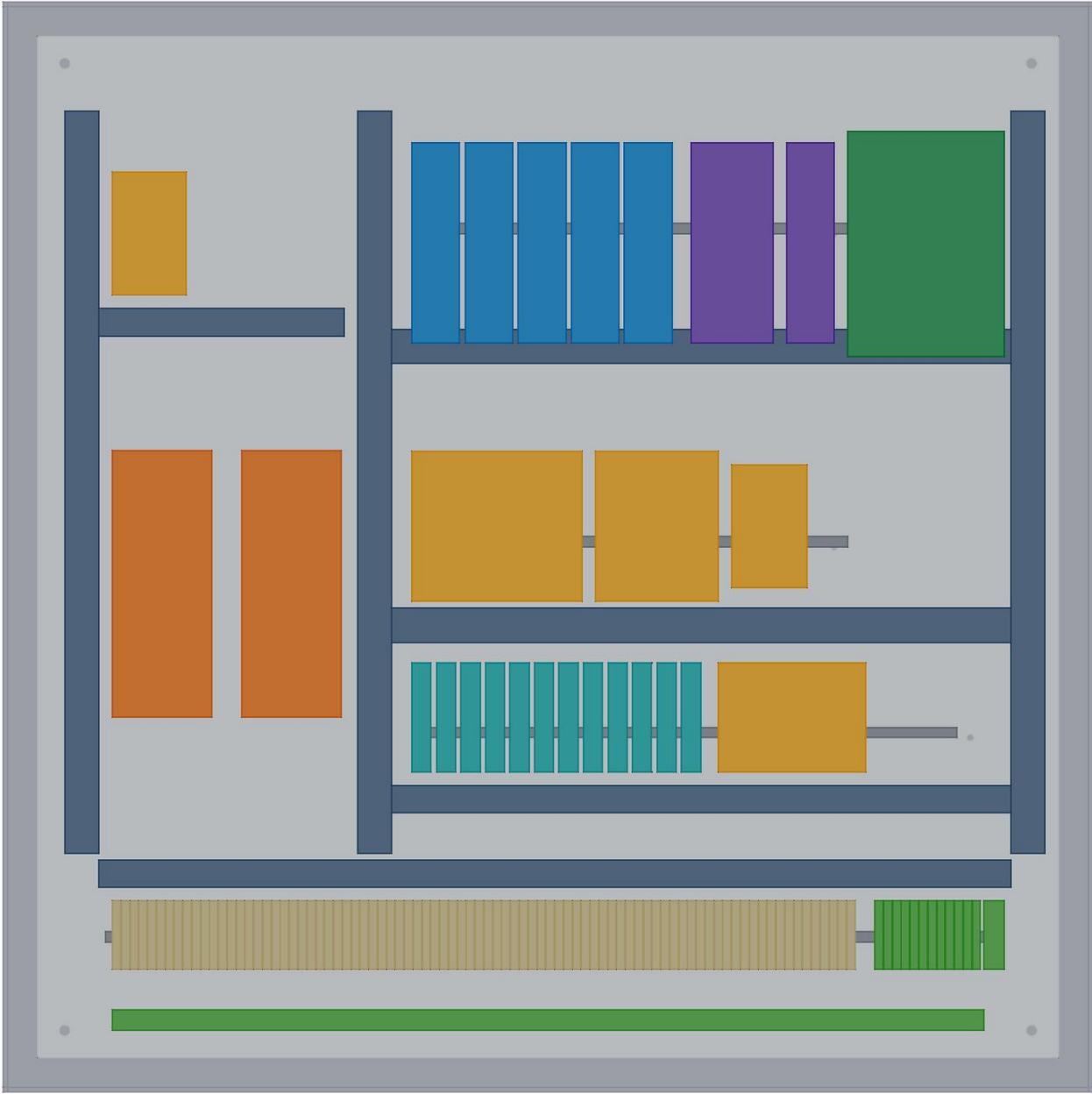




SELECTED / CONTROLLED ENVELOPES

- QF100 54 × 90 × 80 mm [ASSUMED]
Main protective device envelope
- U100 73 × 196 × 225.4 mm [SELECTED]
SINAMICS G120C PN 0.75 kW unfiltered — FSA PN envelope
- U101 73 × 196 × 225.4 mm [SELECTED]
SINAMICS G120C PN 0.75 kW unfiltered — FSA PN envelope
- PS100 125 × 110 × 130 mm [SELECTED]
24 VDC 20 A power supply
- QF110 90 × 110 × 80 mm [SELECTED]
24 VDC electronic branch protection
- K100 55 × 90 × 85 mm [SELECTED]
External conceptual safety/control-enable interface
- A100 35 × 147 × 129 mm [SELECTED]
SIMATIC S7-1500 CPU 1511-1 PN
- A101 35 × 147 × 129 mm [SELECTED]
TM Count 2x24V
- A102 35 × 147 × 129 mm [SELECTED]
DI 32x24V DC HF
- A103 35 × 147 × 129 mm [SELECTED]
DQ 32x24V DC/0.5A HF
- A104 35 × 147 × 129 mm [SELECTED]
AI 8xU/I HF
- SW100 60 × 147 × 125 mm [SELECTED]
SCALANCE XC208 managed switch
- FW100 35 × 147 × 127 mm [SELECTED]
SCALANCE S615 EEC firewall/router
- PC200 115 × 165 × 80 mm [ASSUMED]
Jetson Orin NX industrial carrier reserved envelope
- H100 198.3 × 142 × 55.6 mm [SELECTED]
MTP700 Unified Comfort body

BOUNDARY

Enclosure body 800 × 800 × 300 mm. Mounting plate 750 × 750 × 3 mm. External isolator handle increases full assembly depth to 327 mm. HMI is door-mounted and excluded from the backplate footprint.

FC01 REVISION E — NATIVE FRONTAL INTERNAL ARRANGEMENT

FICTIONAL ENGINEERING PROJECT — NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE — REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.



DEPTH

Selected G120C depth is 225.4 mm; mounting-plate front is Z=18 mm and door inner plane is Z=297 mm, retaining nominal front clearance.

HEAT

Two G120C loss values are controlled at 38.6 W each. PLC, PSU, Jetson carrier, ambient, solar load and enclosure dissipation remain open.

SERVICE

Door opening ≥105° and 1000 mm front service access are controlled assumptions pending site and enclosure confirmation.

REPLACEMENT

Top PLC/network, mid control-power, relay, drive and terminal zones preserve front access; actual connector bend radii and wiring duct fill remain open.

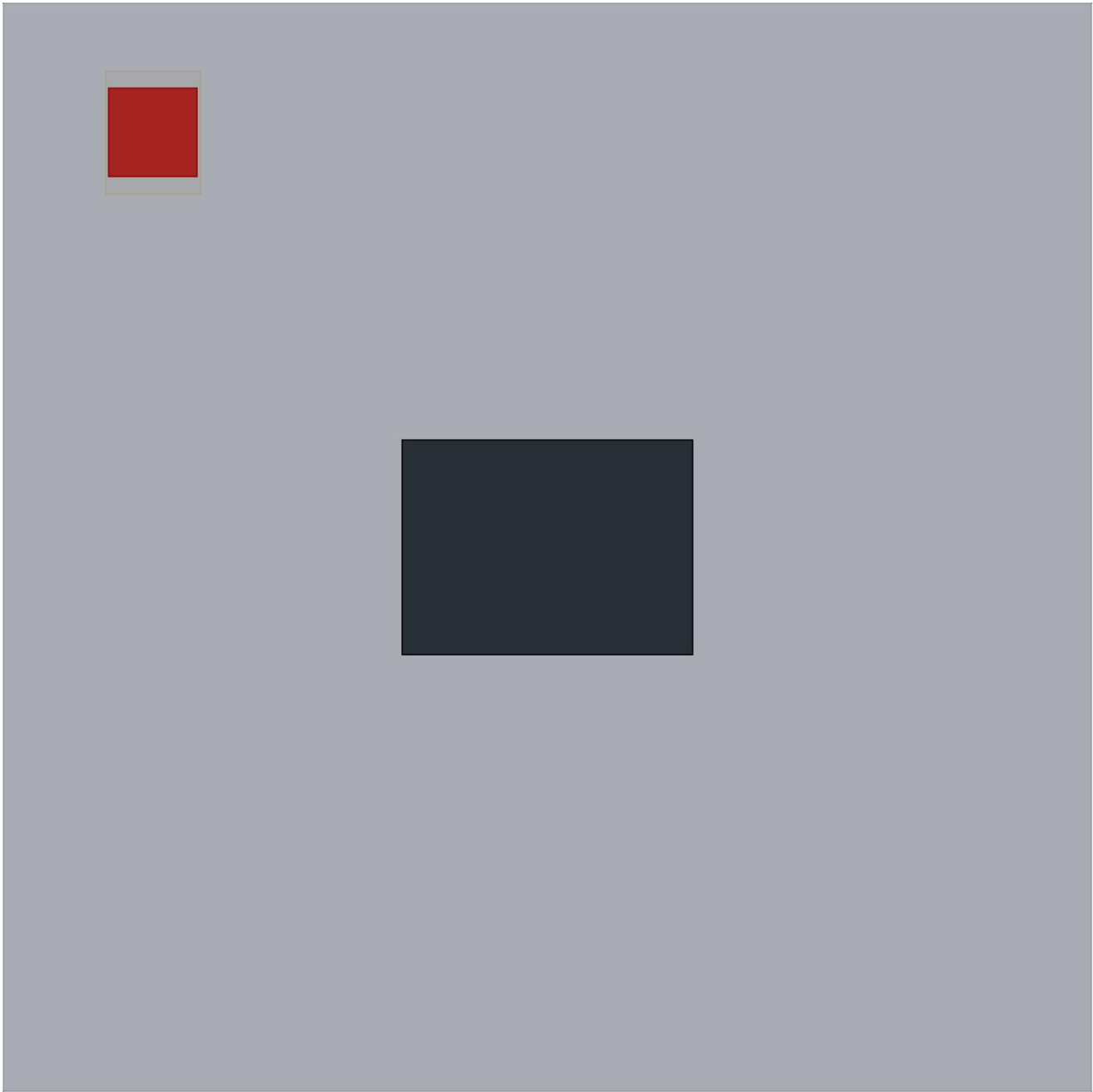
EXPANSION

Right-side/top zones reserve the architecture-only edge envelope and network devices; spare I/O remains controlled in the PLC schedule.

FC01 REVISION E — NATIVE ISOMETRIC INTERNAL ARRANGEMENT

FICTIONAL ENGINEERING PROJECT — NOT FOR CONSTRUCTION

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-H100

MTP700 Unified Comfort outer envelope 214 × 158 × 63.6 mm; controlled cutout 198.3 × 142 mm. Door reinforcement and native WinCC configuration remain open.

-QS100

Door-coupled isolator is a controlled generic envelope. Device rating, short-circuit basis, handle/shaft geometry and regulatory selection remain open.

Door load

Hinge capacity, gland/cable-loop routing, door bond, ingress protection and operator reach require the selected enclosure and site design.

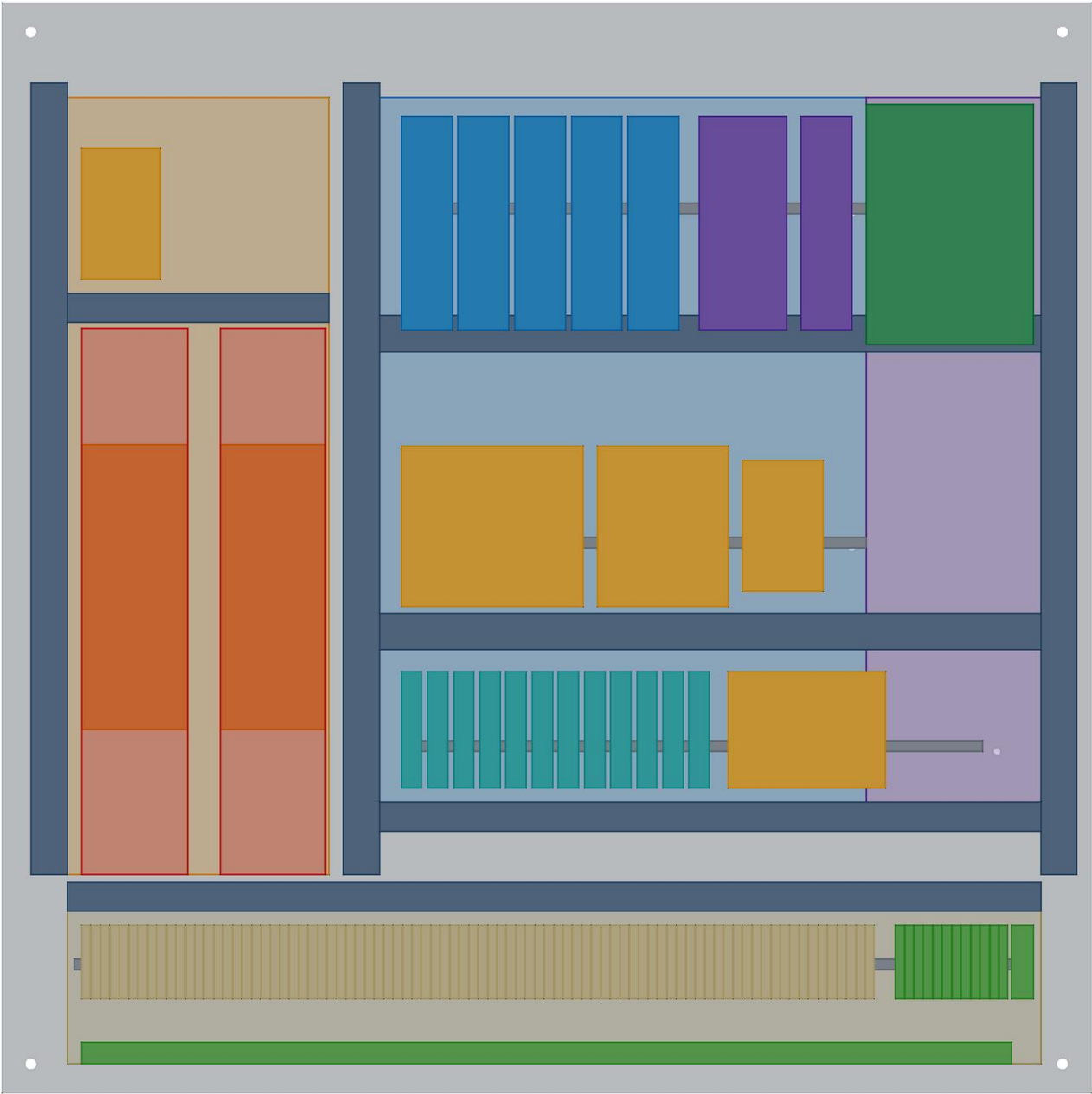
Excluded

No service outlet is included because supply, jurisdiction and selected device are absent. A cooling-device reserve is non-physical only.

FC01 REVISION E — NATIVE DOOR EQUIPMENT ARRANGEMENT

FICTIONAL ENGINEERING PROJECT — NOT FOR CONSTRUCTION

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- Mains / drive heat zone
 - PLC / 24 VDC control zone
 - Network / vision zone
 - Field terminals / PE zone
 - G120C cooling keep-out
- Mains/drive and motor routes remain left of the central segregation duct.
 - Analog/HSC and control conductors are routed away from drive input/output conductors.
 - Ethernet and vision routing remain in the right-side network/vision zone.
 - Shield clamps and PE bar are placed at the field-entry/termination boundary.
 - All duct fill, bend radii, shield hardware, PE continuity and EMC validation remain external gates.

FC01 REVISION E — NATIVE SEGREGATION AND DRIVE-CLEARANCE VIEW

FICTIONAL ENGINEERING PROJECT — NOT FOR CONSTRUCTION

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